ModelIt!

Student Activity Pack

Can We Power a City with Wind?

The Science and Challenge of Renewable Energy for Everyone

Grade 04L2 | NGSS 4-ESS3-1, 4-PS3-1 | Cause and Effect

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Have you ever seen a wind turbine? What do you think it does?

2 If wind is free, why do you think we still use fossil fuels like coal and natural gas for electricity?

3 Draw what you think happens inside a wind turbine when the wind blows.

4 What would happen to a city that gets all its electricity from wind if the wind stopped blowing for three days?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Wind Speed | Number of Turbines | Electricity Generated | City Energy Demand | Grid Reliability

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Windy Day, Plenty of Turbines

Wind Speed at 80%, Turbines at 80%

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Calm Day, Plenty of Turbines

Wind Speed at 15%, Turbines at 80%

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Windy Day, Few Turbines

Wind Speed at 80%, Turbines at 20%

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Battery Storage — Large batteries that store extra electricity from windy days to use during calm days — like saving money in a bank for emergencies

Solar Panels — A second renewable energy source that generates electricity from sunlight — often windy days and sunny days happen at different times

Fossil Fuel Backup — A natural gas power plant that can be turned on quickly when renewables are not producing enough — reliable but produces carbon emissions

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

If wind is free and everywhere, why can we not just use wind turbines to power every city in the world? Use evidence from your simulation to support your answers.

1. **What is the biggest advantage of wind energy? What is its biggest disadvantage?**

2. **Why is it important for electricity supply to MATCH demand at all times?**

3. **How could combining wind energy with solar energy make the grid more reliable?**

4. **If you were a city planner, would you choose 100% wind energy or a mix of sources? Why?**

5. **How did your model help you understand why switching entirely to wind energy is challenging?**

8 STEM Challenge: Design a Wind-Powered Community

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YOUR ENGINEERING MISSION

Design a plan for a community that gets most of its electricity from wind, including a solution for what happens when the wind stops.

THE SCENARIO

A new eco-friendly community is being built and wants to be powered primarily by wind energy. Your engineering team must design the community's energy plan: how many turbines are needed, where they should go, and what backup plan will keep the lights on during calm days. Use your model evidence to justify every decision.

Engineering Design Thinking

1. HOW WILL YOU CALCULATE THE NUMBER OF TURBINES NEEDED BASED ON YOUR COMMUNITY'S SIZE?
2. WHERE IS THE BEST LOCATION FOR YOUR WIND FARM (HILLTOP, COASTLINE, OPEN PLAINS)?
3. WHAT BACKUP ENERGY SOURCE WILL YOU INCLUDE (SOLAR, BATTERIES, HYDROPOWER)?
4. HOW WILL YOU CONVINCE COMMUNITY MEMBERS THAT WIND ENERGY IS WORTH THE INVESTMENT?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Wind Energy Engineers and Renewable Energy Specialists design, build, and maintain wind farms that generate clean electricity. As the world shifts to renewable energy, this field is growing rapidly. They earn $65,000–$120,000/year.

9 Post-Assessment

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Show What You Learned!

Multiple Choice: Now that you have completed the lesson, circle the best answer for each question. Compare your answers to your pre-assessment — did your thinking change?

1 Wind turbines convert one form of energy into another. What energy transformation happens inside a wind turbine?

- **A.** Heat energy from the sun is converted into wind energy
- **B.** The kinetic energy of moving air is converted into electrical energy
- **C.** Electrical energy is converted into wind energy to make the blades spin
- **D.** Sound energy from the wind is converted into light energy

2 In the model, Wind Speed and Number of Turbines are external components. Why are these considered external?

- **A.** Because they are located outside in the wind farm
- **B.** Because they are conditions or design choices that determine how the rest of the system behaves
- **C.** Because they are the smallest parts of the system
- **D.** Because they respond to changes in Electricity Generated

3 The model shows that City Energy Demand has a NEGATIVE relationship with Grid Reliability. What does this mean for a city?

- **A.** When the city uses more energy, the grid becomes more reliable because more power flows through it
- **B.** When City Energy Demand increases but electricity supply stays the same, Grid Reliability decreases because demand outpaces supply
- **C.** City Energy Demand and Grid Reliability always move in the same direction
- **D.** Grid Reliability has nothing to do with how much energy the city needs

4 A student runs the model with Wind Speed at 15% and Number of Turbines at 80%. Grid Reliability drops very low. What does this tell us about wind energy?

- **A.** Wind energy does not work because turbines are too expensive
- **B.** Having many turbines guarantees reliable power for the city
- **C.** Wind energy is intermittent, so even many turbines produce very little electricity on calm days, causing the supply to fall below the city's demand
- **D.** The turbines broke because the wind was too weak to spin them

5 Based on the model, why can wind energy alone NOT reliably power a whole city?

- **A.** Wind turbines are too small to make enough electricity
- **B.** Wind is free, so the city would not make any money from it
- **C.** Wind speed changes constantly, so electricity generation goes up and down while the city's demand stays high, causing blackouts on calm days
- **D.** Wind energy only works at night when people are sleeping